

M003 · First Steps with Coordinates (x, y) — English Worksheet

Topic: First Steps with Coordinates (x, y) · **Course:** 3D Coordinates · **Time:** about 45 minutes

This lesson you draw on a flat **wall**. Every spot has a name — **two numbers (x, y)**: x is how far **across**, y is how far **up**, counting from **0**.

1 · Predict 🧠

Read each set of steps. Write what you think you will see.

```
place gold block at (3, 0)
place gold block at (3, 5)
```

The two gold blocks share the same x. Are they close together or far apart? Which number changed?


```
place red block at (0, 0)
place red block at (4, 0)
place red block at (0, 4)
```

Three red blocks. Which one sits at the corner where counting starts? Which one is higher up the wall?


```
set y to 2
place gold block at (0, y)
set y to 6
place gold block at (0, y)
```

The variable y changes from 2 to 6. Where does each gold block land? Do they share the same x ?

2 · Spot the Bug 🐛

Each code block below is broken. Read what it should do, fix it, then explain why the original was wrong and your fix works.

Bug A — The block should land at **(2, 5)** — 2 across, 5 up.

```
place gold block at (5, 2)
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The block should sit at the **bottom corner** of the wall. The coder forgot where counting starts.

```
place gold block at (1, 1)
```

Hint: counting starts at **0**, not 1.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — The block should sit **5 up** at $x = 0$ — that is **(0, 5)**. It drops below the wall instead.

```
place gold block at (0, -5)
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

2 · Your build. Point the camera. Name the parts.

Today I built _____.

I built it using this code: ____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

The most fun part was _____.

Something new I learned was _____.

Write your lines here, then say them in your video.

Submit 

Send this worksheet + your video to teacher on KakaoTalk.